

The Model Store a certified retrieval machine

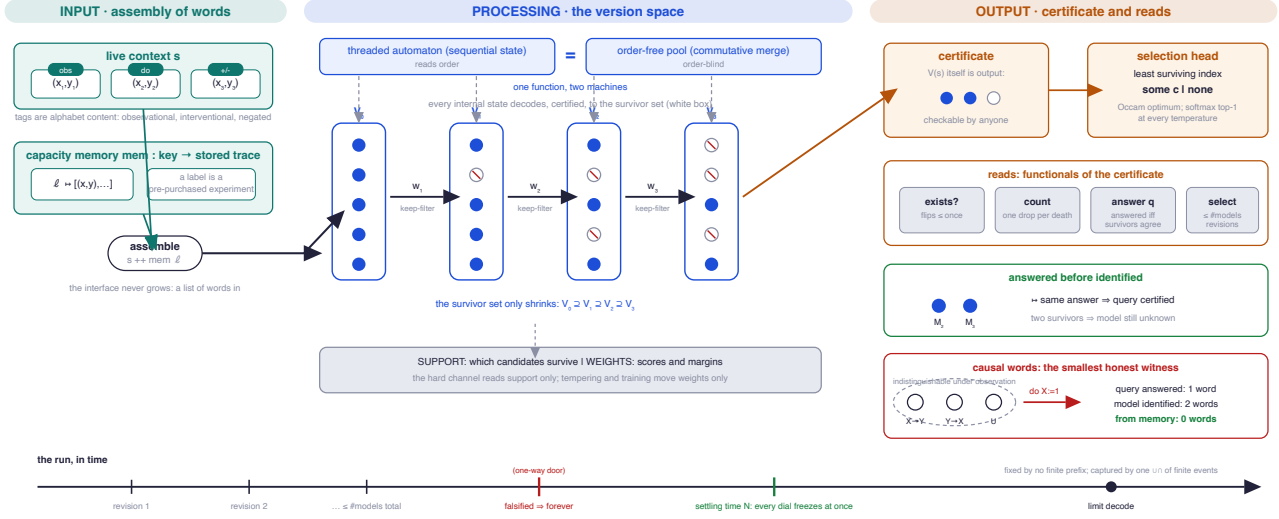
A constructed object. Every property below is a machine-checked theorem with exact constants; the stated instances execute inside the proof checker.

The object. A finite list of candidate models $c_1, \dots, c_k$, each a program $Dc : X \rightarrow Y$, operated as a retrieval machine over observation words.

INPUT. A context: a finite list of words (x, y) , assembled from live evidence and from memory (a label is a key whose stored trace is appended to the context). Tags on words carry observational, interventional, and negated readings. The interface is this one list type; it never grows.

PROCESSING. Maintain the *version space* $V(s)$: the candidates consistent with every word so far. One observation is one keep-filter; the space only shrinks. Computed by a sequential state machine or by an order-free commutative merge, which agree on every context. Every internal state decodes, with a checked certificate, to $V(s)$.

OUTPUT. The least-indexed survivor (or *none*), together with $V(s)$ itself as the certificate. All further answers are reads of the certificate: existence, counts, and per-model query functionals, each certified against every survivor.

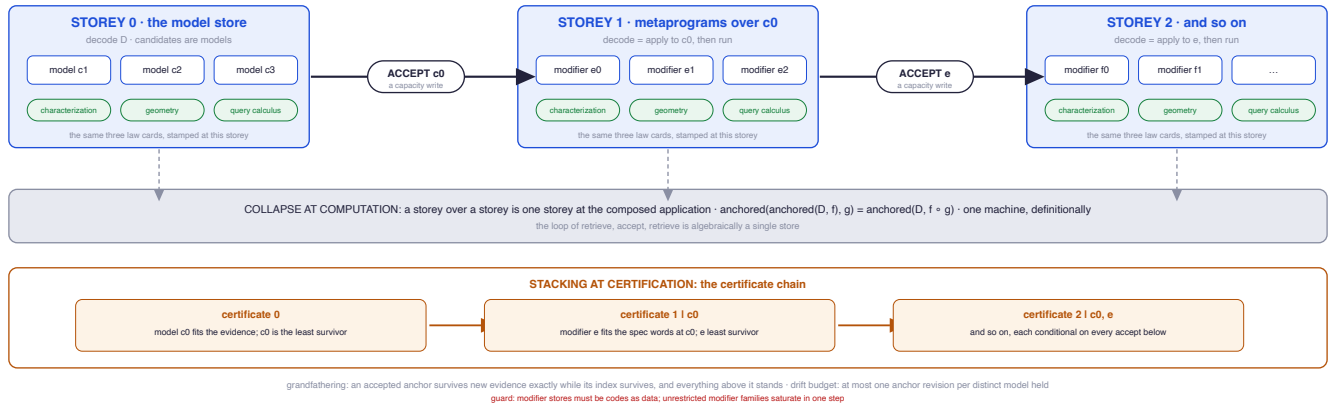


PROVEN PROPERTIES

- P1. One function, two machines.** The threaded state machine and the order-free merge compute the same answer on every context, for every store, with no side condition.
- P2. White box.** Every reachable internal state decodes to the survivor set, and any two faithful decodings agree along the run: there is no hidden internal freedom.
- P3. Exact capacity price.** The smallest machine reproducing the store's answers has size exactly the number of reachable survivor configurations. Sequential state count and parallel width times precision are the same price in two currencies, exchanged by complementation.
- P4. Order is a resource.** Order-free processing recovers everything determined by the survivor set and provably cannot recover the order of eliminations; a sequential pass recovers it exactly.
- P5. Exact revision budget.** The answer changes at most once per distinct model held, and an adversarial data order spends the whole budget. Duplicate encodings of one model never cost a revision.
- P6. Universal revision inheritance.** The survivor set itself changes at most k times, and every read of it inherits that budget. The existence read flips at most once: a refutation is permanent, while consistency can always be overturned by later evidence.
- P7. Temperature-free Occam selection.** The output is at once the least-indexed survivor, the optimum under every complexity-decreasing weighting, and the softmax top choice at every temperature. Hardening the soft regime moves the weights while the answer stays fixed.
- P8. Scores are independent state.** The weight stratum (scores, margins) is not fixed by the survivor trajectory: two runs with identical survivor sets at every step can carry different scores. This is the soft channel's dark knowledge, all it exposes beyond the answer, and it is a function of the candidates that lost.
- P9. Simultaneous settling.** On any unbounded stream the survivor set is eventually constant, and every readable dial freezes at one common time.
- P10. Identification at the limit.** If some candidate fits the whole stream, the limit answer is a candidate consistent with every prefix, reached within the revision budget. A target outside the list is approximated by the simplest consistent candidate; identification requires the target to be in the list, which is the machine's only boundary.
- P11. The limit, exactly bracketed.** No finite context determines the limit answer, yet the limit event is fully captured by one countable union of countable intersections of finite-context events.
- P12. Answered before identified.** A query is answered the moment all survivors agree on its value, with a certificate, possibly long before the model is pinned; answering can be strictly cheaper than identifying.
- P13. Labels are pre-purchased experiments.** A label's whole effect is the intersection law of its stored trace. It cannot separate candidates its stored words cannot kill, and a query can cost zero live words when the right experiment is already in memory.
- P14. Evidence of every polarity is alphabet content.** Demands to fail on a word, and words read under intervention, enter by recoding the alphabet; every property above transfers to the recoded store unchanged. On the smallest causal store, mechanisms indistinguishable under observation are separated by one interventional word: the query is answered at price one, the model identified at price two, and the same query costs zero from memory.

The Second Storey metaprogram retrieval, and the limits of finiteness

The argument. A retrieved model can be accepted as the *anchor* of a next store, whose candidates are modifiers: codes that apply to the anchor and run. The anchored decode is a decode, so a store of modifiers is a store, and retrieval one storey up is the same machine. The query “here is my model, retrieve the modifier meeting my new conditions without touching the structure I accepted” is ordinary retrieval at the second storey; the loop of retrieve, accept, retrieve closes into a cascade.



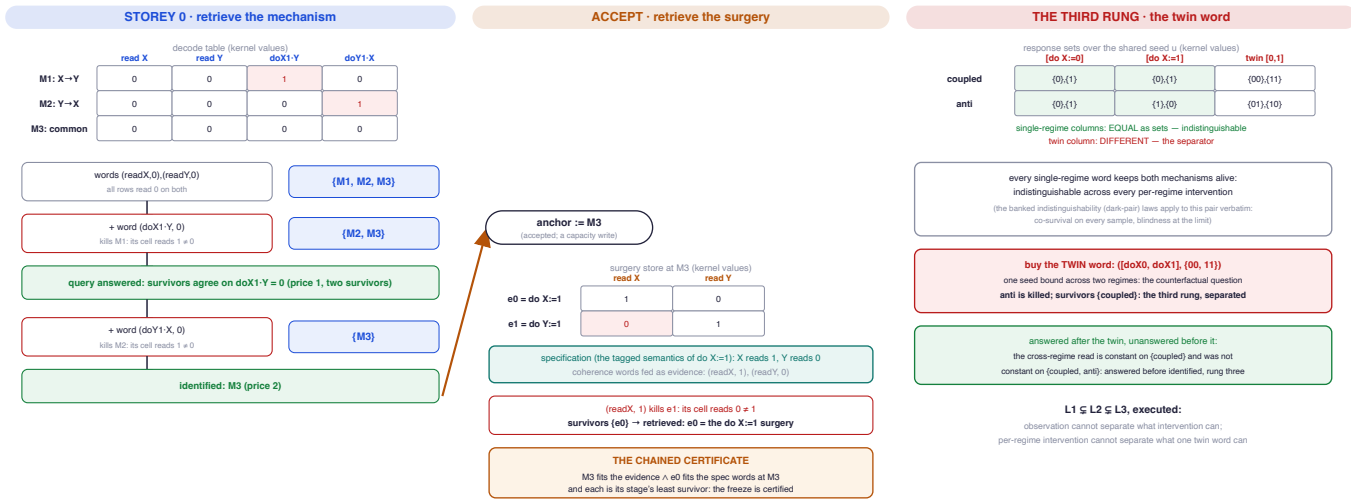
- M1. Laws at every storey.** Every anchored store, for every anchor and every family of modifiers, carries the complete characterization, the full geometry, and the whole query calculus of pages one and three. The analysis of the machine is storey independent: one theorem covers all levels at once.
- M2. Collapse at computation, stacking at certification.** Composing anchorings is, definitionally, one anchoring at the composed application: the cascade is a single store at the level of computation. What stacks is the certificate chain: each acceptance freezes an anchor, and every downstream certificate is conditional on the accepts below it. This is the collapse test of page six, applied to the cascade.
- M3. The chained certificate carries the freeze, at every depth.** A cascade of any length is certified at every storey: each stage’s modifier fits its evidence at its anchor and is that stage’s least survivor, each certificate conditional on the accepts below it. Running one cascade after another is running their concatenation; the two-stage certificate is the single-storey reading. The greedy selection is itself part of the certificate.
- M4. Grandfathering, with an exact drift budget.** An accepted anchor survives new evidence exactly while its index survives, and while it does, everything built on it stands unchanged. The anchor itself is revised at most once per distinct model held: the model-grain revision law, read at the base store.
- M5. Correctness is evidence.** A modifier meets a specified behavior on given inputs exactly when it fits the words assembled from that specification. Checking a candidate surgery against its specification is a run of the machine on specification words, and the resulting certificate says so: its content is agreement with the specification, and that alone.
- M6. Where finiteness ends.** At a countably infinite store, identification on realizable streams survives (the ordering of the indices replaces the finite count), and the per-word laws survive with it. What fails is exactly the uniform part: no store-size bound on revisions exists, unconditional settling fails off realizability, and selection becomes specifiable but not computable. Finiteness buys the uniform, exact laws; identification is bought by the ordering alone.

Modifier stores obey the machine’s own guard: candidates are codes as data (unrestricted modifier families collapse the storeys in one step; grammar-bound modifiers climb, page four). Retrieval over level-sorted modifiers returns the least-level fit; anchoring adds nothing to the selection price, and execution pays a per-storey capacity product at every depth, with the measurability side discharged on standard domains. Adjacent frameworks (adapter fine-tuning, library learning, partial evaluation, knowledge compilation, specification monads) share the shape; none certifies its retrieval or prices it exactly.

The Witness, by Construction

one causal question through every storey

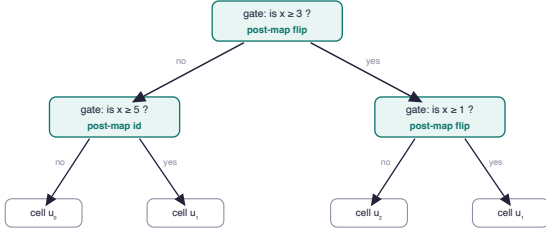
Every table cell and every survivor-set transition in the figure below is evaluated inside the proof checker. The figure exhibits the argument: it is evaluated inside the checker, so reading the tables and checking each elimination by hand reproduces the proofs. Three rival mechanisms remain indistinguishable under observation; one correctly chosen intervention answers the pressing question while two candidate mechanisms remain consistent; the second intervention identifies the mechanism; the accepted mechanism anchors a store of surgeries from which the requested modifier is retrieved on specification words, with a joint, conditional certificate; and two mechanisms agreeing under every separate intervention are told apart only by the twin question that binds one hidden seed across two regimes.



The Witness the router forest, as built

The realization. The machine of page one is abstract: any candidate list, any decode. The witness is the concrete object it was discovered inside: a routed network read as a *forest of program trees*. A candidate is a finite tree; a gate at each node chooses a child by a threshold on the input; an elementary compute cell sits at every base and a finite post-map at every node, so the network's compute family is constitutive to the grammar. The machine's retrieval runs over this forest verbatim, and everything on this page is a theorem about the drawn objects.¹

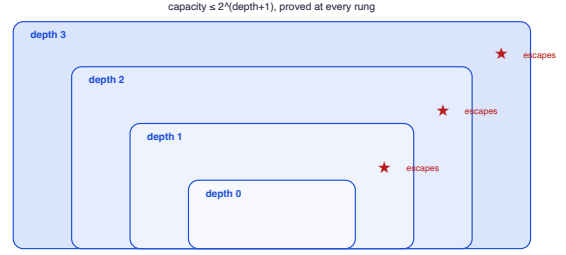
A PROGRAM IS A TREE · drawn from the construction



one candidate = one finite tree: a gate picks a child by a threshold on the input, the chosen child answers, the node's post-map finishes. The compute family is constitutive: an elementary cell at every base, a finite post-map at every node.

free selection is forbidden by the machine's own theorem: with unrestricted selectors, one step reaches every function, so candidates must be trees (data).

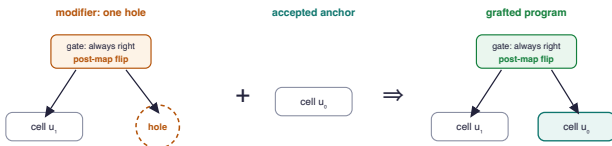
THE TOWER · strict at every rung, priced at every rung



every rung strictly larger, on the counting world and on the real-vector world; the union over all depths never exhausts the function space. Retrieval over depth-sorted candidates returns a least-depth fit: built-in simplicity bias.

the machine of page one runs verbatim over this forest: same retrieval, same laws.

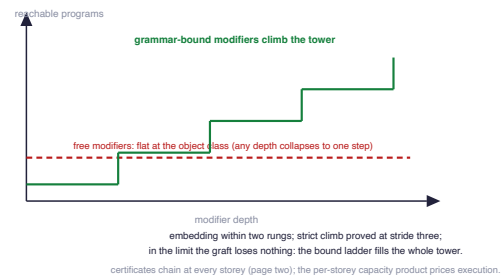
MODIFY BY GRAFTING · the first rung climbs, in one picture



grafting is part of the grammar: a plugged modifier is a tree, so modified programs inherit every law. Here the gate always routes to the graft and the post-map flips it: the grafted program disagrees with its anchor on every input. One modifier, one rung.

retrieval over modifiers = grafting: the accepted anchor enters every candidate by the hole.

FREE IS FLAT, BOUND CLIMBS · the dichotomy



- W1. Restriction is forced.** If selection over the compute family were a free function of the input, one selection step would already realize every function, and no hierarchy could exist. Restricting selection to gates over tree-shaped candidates is what gives depth its meaning.
- W2. The tower is strict at every rung, twice.** Depth- $\ell+1$ trees realize strictly more than depth- ℓ trees, at every ℓ , on both worlds; one reflection step over any rung escapes that rung; and the union over all depths still never exhausts the function space.
- W3. Capacity is graded by depth.** The depth- ℓ class shatters no set of size $2^{\ell+1}+1$: the price of a rung is known before it is bought, and retrieval over depth-sorted candidates returns a least-depth fit, so the machine's selection is a simplicity bias with a certified price tag.

- W4. Modifiers live in the same grammar.** A modifier is a tree with one hole; applying it to an accepted anchor is grafting; a grafted modifier is again a tree. Retrieval at an anchored store of modifiers *is* grafting, and each modifier frame carries its own post-map and cells, so the compute joint is present at every storey.
- W5. The graft is cheap and loses nothing.** Everything at depth ℓ reappears among grafted modifiers within two rungs; the grafted ladder provably climbs (strictly within every three rungs, and already at its first rung); and in the limit the grafted ladder reaches exactly the whole tower.
- W6. The dichotomy.** Free modifier families are flat: any depth of anchoring reaches exactly the object class, because composed anchorings collapse to one. Grammar-bound modifiers climb the tower. Depth is a resource exactly when modifiers are data in the object's own syntax.

Still open: whether the two-rung graft embedding and the three-rung strict climb are sharp (the drawn one-rung climb is proved at the first rung), and whether the threshold gates read literally as scored routers, which would let the per-storey price law of page two apply to the tower word for word. Every drawn tree, gate, and graft above is a term in the verified construction.

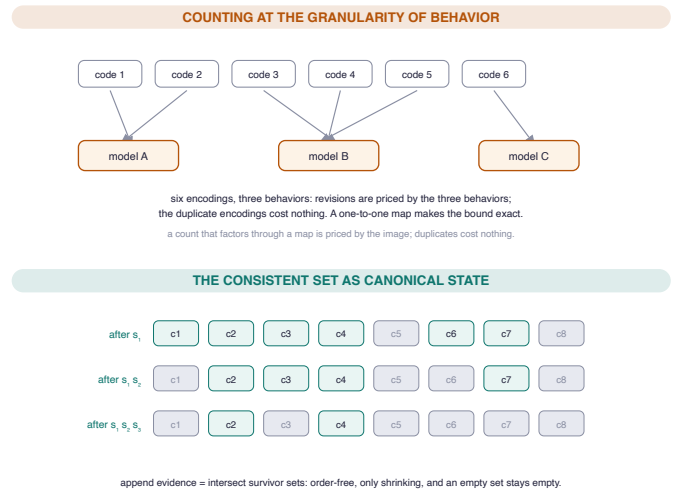
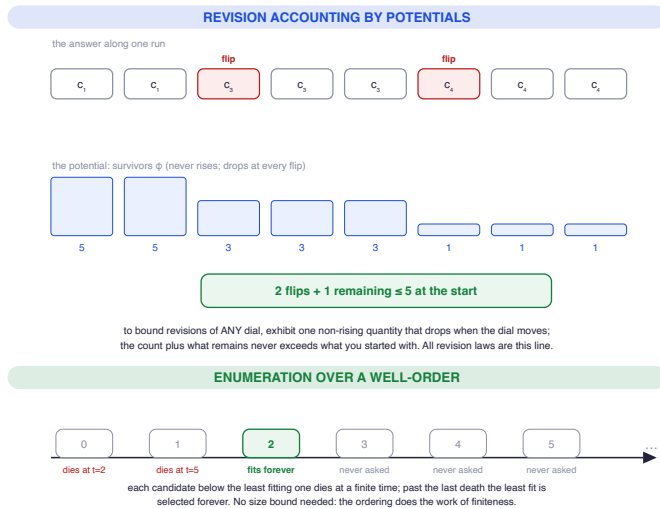
¹Capacity grading is a Sauer-Shelah-type bound per rung; the strict hierarchy is proved on the counting world and on the real-vector world with affine threshold gates. The saturation theorem that forces tree-shaped candidates is a Cantor-style diagonal argument at the selection layer.

Methods, I ways of counting

The machine's laws are instances of a small number of free-standing ideas. Each is stated here as its own tool, usable on any system, with the machine as one worked instance; each is a verified general theorem whose machine instances follow as corollaries.

Revision accounting by potentials. How often can a running process change its output? Ad hoc answers reprove one induction each time. Pair the observed dial with a potential: a quantity that never increases and strictly decreases whenever the dial moves. Then the number of moves plus the remaining potential never exceeds the starting potential, so a revision bound is exactly an exhibited potential. In the machine the survivor count is one potential serving every dial at once, which is why the answer's budget, the mask's budget, and every read's budget are a single argument.²

Counting at the granularity of behavior. A count stated over a list of encodings scales with the list; but a list over-counts, since distinct entries can encode equal behaviors. Any count that factors through a map is bounded by the size of the image: the number of distinct values it takes, however many inputs share them. The bound becomes an equality when the map is injective. Prices then attach to the set of distinct behaviors, indexed by what each candidate does and blind to how it is encoded: for the machine, revisions are priced by distinct models, and the price is exact once duplicate encodings are quotiented away.



The consistent set as canonical state. What must an incremental learner remember? Maintain the set of candidates consistent with everything seen so far. Appending evidence intersects these sets, in any order, and an empty set absorbs. This set is a canonical state in a precise sense: a sequential implementation and an order-free one agree because both decode to it, and its reachable configurations lower-bound the size of any faithful implementation. Every answer the machine gives is a read of this one shrinking set.³

Enumeration over a well-order. Finite-class convergence arguments spend the class size; an infinite class offers none. Order the candidates, and suppose some candidate fits the whole stream. Every earlier candidate is refuted at a finite time, so past the last such refutation the least fitting candidate is selected forever: the order supplies what the count did. The separation is the content. Convergence survives the move to an infinite store, since the order still does its work; the uniform budgets do not, since they were the one thing finiteness was buying.⁴

Each method is one verified statement with its machine instances derived as corollaries; the figures carry worked numbers, so checking the arithmetic in a panel reproduces the corresponding proof's content.

²The count of output changes along a stream is mind-change complexity: E. M. Gold, *Language identification in the limit*, 1967; J. Case and C. Smith, *Comparison of identification criteria for machine inductive inference*, 1983.

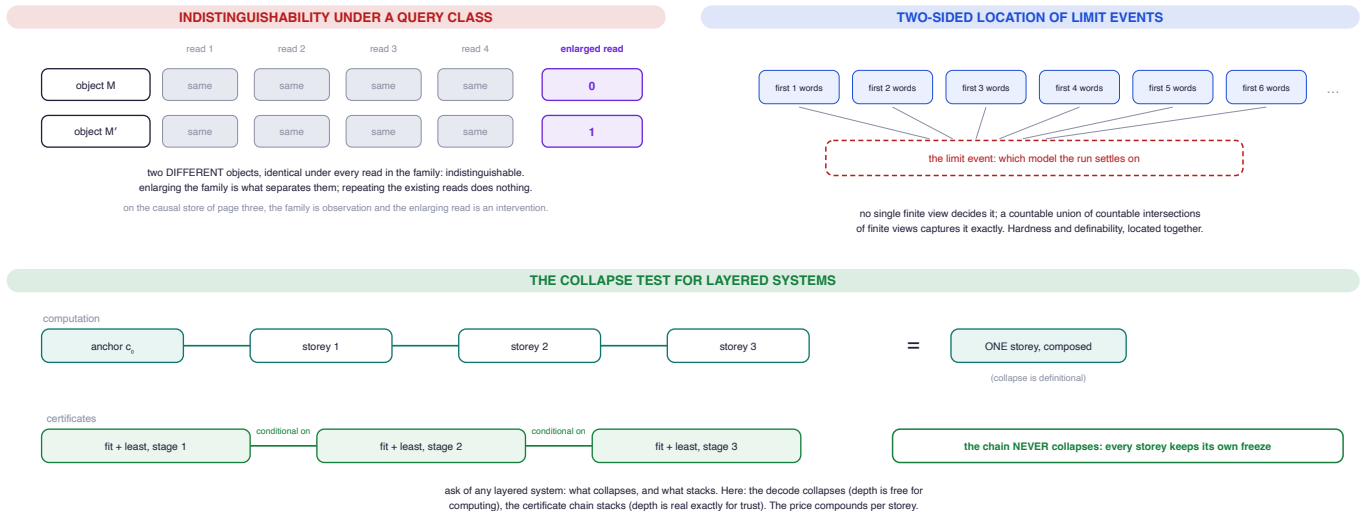
³The consistent set is the version space: T. M. Mitchell, *Generalization as search*, 1982.

⁴Identification by enumeration: E. M. Gold, 1967. The infinite-store positive and its two negative companions are page two's final entry.

Methods, II ways of separating

Indistinguishability under a query class. Fix the class of queries an analyst may put to a system. Two hypotheses that agree on every query in the class cannot be separated by any procedure restricted to it, however long it runs; call such a pair indistinguishable relative to the class. The indistinguishable pairs are a complete inventory of what the class can never determine, and each pair prices its own resolution: the smallest enlargement of the class by a query on which the pair disagrees. Under observational queries on the causal store the indistinguishable pairs are the Markov-equivalent mechanisms and the enlarging query is an intervention, whose word price the machine computes; inside the machine, weights are indistinguishable relative to survivor-set queries.⁵

Two-sided location of limit events. Whether a process has settled is decided by no finite amount of data; alone, that is a hardness remark with no upper bound. Locate the event instead: no finite prefix decides it, and it equals a countable union of countable intersections of prefix events. The two sides priced together are the result: a hardness floor (no finite view suffices) and a definability ceiling (this level of view does), which makes the limit verifiable in the tail while remaining undecidable at every finite stage.⁶



The collapse test for layered systems. Is depth a resource? Compose two layers and ask the question separately at each register the system exposes. Where composition is definitionally one layer, depth carries no content in that register; where composition refuses to collapse, depth is real there. For the cascade the answer function collapses (any tower of anchorings is one anchoring) while the certificate chain does not (every storey keeps its own conditional freeze), and the statistical price compounds per storey. So depth is free for computation and real for trust and capacity, and the test says exactly where depth is worth spending, sharpened by page four: it becomes expressive when modifiers are bound to the grammar.⁷

Modification inside the object grammar. Editing an accepted object usually spawns a separate apparatus whose relation to the base theory has to be re-established. Instead define modifiers as the object's own syntax with one hole, and let application graft the accepted object into the hole. Because grafting is an operation of the grammar, a modified object is again a grammatical object: every law of the class transfers with no new theory, the modifier class has a capacity measured in the same grammar, and modifications compose. This is what makes retrieval over modifiers ordinary retrieval, and what restores the compute family to every storey of the cascade.⁸

The frontier. Stated as obligations, in the object's own terms: the sharpness of the graft prices; the scored reading of the gates; the distributional layer over the deterministic mechanisms; and the exact price law of experiments. Each names structure still to be built; every law above stands.

Everything on these six pages is machine-checked in Lean 4 over Mathlib. Axiom footprint: the standard triple or less, several central laws axiom-free; no unfinished proofs, warnings promoted to errors. The executed instances (pages three, five, six) run inside the checker's decision procedure: the tables and worked panels are evaluated there.

⁵On causal graphs, observational indistinguishability is Markov equivalence: J. Pearl, *Causality*, 2009.

⁶For the connection between revision counts and this definability level: R. Freivalds and C. Smith, *On the role of procrastination in machine learning*, 1993; M. de Brecht and A. Yamamoto, 2010.

⁷The collapse is Kleisli composition of the retrieval steps: E. Moggi, *Notions of computation and monads*, 1991.

⁸One-hole contexts and their calculus: G. Huet, *The zipper*, 1997; C. McBride, *The derivative of a regular type is its type of one-hole contexts*, 2001.